

Functional Neuroimaging

NEUROSCI 382, PSY 303

Fall 2026, Tuesday 3:05-5:45 PM, Perkins LINK 088 (Classroom 4)

Prof. Tobias Overath, Hector Sanchez Melendez (TA)

This class introduces you to the world of functional magnetic resonance imaging (fMRI). The **main aim** of the class is to make you thoroughly familiar with, and fascinated by fMRI as a research tool. This will be achieved both through **gaining theoretic (textbook) knowledge** as well as through **hands-on experience** by designing, running, and analyzing your own fMRI study!

Outline of the class:

The class is roughly structured into a **lecture part** and a **lab part**. There is typically a **reading assignment** for each lecture, which you will then consolidate further in the subsequent lab.

The **first few weeks** will cover the basic **physics underlying MRI** and the physiological basis of the so-called **blood-oxygenation-level dependent (BOLD)** signal, which is the mainstay of fMRI research. Following **Midterm I**, you will learn about **study design** and various levels of **data analysis**, before we turn to applications in **clinical populations** (followed by **Midterm II**). More importantly, you will put all of this into practice by **acquiring your own data**.

Group projects:

The **group projects** are a significant part of the class. You will form **groups of four students** each and each group will then – with my guidance – **design a research study, program the experiment and run it in an fMRI scanner**, and finally **analyze the data**.

Each group is assigned **two 2-hour scanning sessions**, so you can scan 2 participants. Note that each hour of scanning would normally cost **\$640**, so it is imperative that everything works smoothly (i.e., all scripts are running etc.) before you run your first participant! In the past, students have asked friends or volunteered themselves to be participants. You then need to be present when your participants are doing the experiment in the scanner. There will be **class periods that are dedicated for analyzing your acquired fMRI data**.

Two students from each group will **present an outline** of the project in Week 10 (i.e. before you've acquired any data), while the other two students will **present the results** in the last class in Week 15. Each student will then submit an **individual write-up** on their project (~3000 words) on the **Tuesday, December 8, of Finals Week**.

Textbook:

As a **textbook**, we will use the third edition of “**Functional Magnetic Resonance Imaging**” by Scott Huettel, Allen Song, and Gregory McCarthy (2014, Sinauer). The Duke Bookstore should have copies for sale and it's available online.

The class also has a **site on Canvas**, where all the **reading materials** (other than the textbook) will be posted under **Files** for each class in advance. I will also post the lectures on Canvas for you to review after the class. Please **submit your write-up** of your project in Canvas and directly via email.

Reading assignments:

Most classes have **assigned reading** materials that can be quite dense and long. It is **important that you read this before coming to class!** This will help you digest the material when we go through it in class.

This is a small class, so your impact on it is potentially much larger than in a large and established lecture. If you have suggestions about topics that you find interesting and that should be covered, please do not hesitate to let me know.

This is also a **unique class** in that you get to acquire **your own fMRI data** and analyze it! I hope that – when you look back at your undergraduate years – this class will stand out and that you will remember it as one of your most rewarding classes during your college years.

I should also add that the class at times requires a **considerable time commitment** on your part. The textbook readings are often **dense** and the first few weeks in particular – when we are covering the basics of MRI and fMRI – can be challenging. Furthermore, **experimental science is time consuming** and often involves the interference of unforeseen stumbling blocks, so your group project will have phases that require considerable time commitments. The idea, however, is that the **thrill and rewarding aspects** of running your own study and analyzing your own data will far outweigh the labor involved. Finally, **the TA and I are here to help**, so don't hesitate to get in touch with us!

Use of technology:

Active use of laptops and other technology (e.g., cell phones, tablets, etc.) is not permitted during the lecture. This policy is based on careful analysis of the scientific literature, which shows that using a laptop in class is associated with poorer grades for both laptop users and for their neighbors.

If you feel strongly that not using a laptop or similar device will impair your learning in this class, please reach out to me to discuss your needs. Anyone using a laptop or similar in class must have instructor approval and a signed agreement indicating that the laptop or similar will be used exclusively for notetaking.

I understand that “life happens”. If you are expecting an important message during class that requires you to check your phone, talk to me beforehand; in that case, feel free to step out for brief periods to check texts, emails, etc.

Student integrity and attendance:

Student integrity is assumed and the **Duke Community Standard** applies (<http://studentaffairs.duke.edu/conduct/about-us/duke-community-standard>). I strongly encourage you to discuss the topics covered in the reading materials and to communicate with each other outside of class! In fact, this is absolutely necessary for the group projects!

However, please be aware that any work that you submit should be your own; if there are aspects that emerged from group discussions or from some other external source, be sure to state this explicitly. **If you are unsure, check with me first!** Should a student violate aspects of the Duke

Community Standard, he/she will either fail the particular graded assignment, or he/she will fail entirely, depending on the severity of the violation.

Attendance is mandatory. If you are unable to attend a class and/or to complete a graded assignment because you are sick, please follow these guidelines for **Short-Term Illness Notification Forms** (STINFs, <http://trinity.duke.edu/undergraduate/academic-policies/illness>); please note the definition of 'incapacitation' ;-)

If you are unable attend a class and/or to complete a graded assignment out of **religious reasons**, please follow these procedures:

<http://trinity.duke.edu/undergraduate/academic-policies/religious-holidays>. If you are a **varsity athlete** and are unable to attend a class and/or to complete a graded assignment, please follow the procedures outlined here: <http://trinity.duke.edu/undergraduate/academic-policies/athletic-varsity-participation>

You must submit a STINF or other notification via email for every class that you have missed for any reason before the end of the day of that class. Each failure to do so will result in subtracting 2% from the total final grade, up to a limit of 10% grade adjustment.

Note that submitting a STINF is a courtesy notification, not a guaranteed accommodation. Instructors are not obligated to approve late submissions, make-up exams, or deadline extensions solely due to submitting a STINF. All accommodations are handled on a case-by-case basis, at the instructor's discretion.

Since the class is taught in two consecutive periods on the same day, missing a day is like missing two classes. Should you miss more than 5 classes (out of 14), you will have missed a total of 10 days of a regular TU/TH class schedule; in this case, unless there are extenuating circumstances and/or you can convincingly demonstrate that you will be able to make up the missed material in a timely fashion, you will not be able to pass this class.

Finally, it is **your responsibility to back up your data!** If, on the day of submitting your final group project paper, you realize that 'the file got corrupted' or 'didn't save properly', then there is nothing I can do about this. The same applies if the file you submit (on time) is corrupted or cannot be graded for any reason. It is your responsibility to ensure that the work you submit can be graded.

I will deduct letter grades for late submissions (1 letter grade for late submissions, and then an additional letter grade for every 6 hours that your submission is late), as this would otherwise not be fair to students who submitted theirs on time.

AI policy for the Project write-up:

Students may not use generative artificial intelligence (AI) tools to support their writing on this assignment. Basic editing and grammar tools are permitted (e.g., Grammarly grammar checker), but tools with AI assistance that can write or revise your writing for you (e.g., standard AI-enabled version of Grammarly, ChatGPT) are not allowed. As a general rule of thumb, tools that offer sentence-level suggestions to correct grammar or improve conciseness (like Microsoft Editor) are acceptable, but AI tools that can generate drafts or rewrite large sections of text are not. For resources on citation conventions, visit plagiarism.duke.edu and library.duke.edu/research/citing. For more information about our AI policy, visit dukepsy101.info/course-policies/integrity. If unsure what is expected or acceptable, ask me or the TA.

Grading basis:

Midterm I: 30%; Midterm II: 30%; Labs & Quizzes: 10%; Project: 30%. Grading will take all of these into consideration for the final grade.

ARC Information:

The Academic Resource Center (ARC) offers free services to all students during their undergraduate careers at Duke. Services include Learning Consultations, Peer Tutoring and Study Groups, ADHD/LD Coaching, Outreach Workshops, and more. Because learning is a process unique to every individual, we work with each student to discover and develop their own academic strategy for success at Duke. Contact the ARC to schedule an appointment. Undergraduates in any year, studying any discipline can benefit!

211 Academic Advising Center Building, East Campus – behind Marketplace
arc.duke.edu • theARC@duke.edu • 919-684-5917

Contact details:

I can be reached multiple ways during the semester:

Tobias Overath:

Office: LSRC building, Room B248A

Office hours: Tuesday 08:30-09:30 AM (and by appointment)

Email: t.overath@duke.edu

Office Phone: (919) 684 6146

Hector Sanchez Melendez (TA):

Office: M057 Conference Room, DIBS Cube

Office hours: Thursday 2-3 PM (and by appointment)

Email: hector.sanchezmelendez@duke.edu

Syllabus

Week 1

1. TU 25 Aug

Intro: Chapters 1&2

Tour of BIAC

Week 2

2. TU 1 Sep

Chapter 3: MR Physics (focus on Conceptual Path)

Lab: Intro to Neuroanatomy

Week 3

3. TU 8 Sep

Chapters 4&5: MR Image formation & MRI contrast

(Chapter 4: read both Conceptual and Quantitative Paths, but 'ignore' equations; Chapter 5, exclude "Motion Contrast", "Signal recovery..." and "Parallel imaging")

Lab: k-Space, Image Contrast

Week 4

4. TU 15 Sep

Chapter 6: From neuronal to hemodynamic activity

Chapter 7: BOLD fMRI

Projects brainstorming

Week 5

5. TU 22 Sep

Chapter 8: Signal, Noise & Preprocessing

Lab: Preprocessing

Week 6

*****This week see me during office hours to discuss your group project!*****

6. TU 29 Sep

MidTerm I

Project planning session

Week 7

7. TU 6 Oct

Chapter 9: Experimental Design

Project planning session

Week 8

8. TU 13 Oct

FALL BREAK

*****This week see me during office hours to discuss your group project!*****

Week 9

9. TU 20 Oct

Chapter 10: Basic Statistical Analyses

Lab: 1st level Analysis

Week 10

10. TU 27 Oct

Chapter 11: Advanced Statistical Analyses

Student group project proposal presentations

Week 11

*****You should start acquiring data as of this week*****

10. TU 3 Nov

How to Write a Paper (Bem: Writing the Empirical Journal Article)

Lab: 2nd level Analysis; if necessary: more Task Programming

Week 12

11. TU 10 Nov

Lab: Project analysis

Week 13

*****You should have acquired all your data BEFORE class this week*****

12. TU 17 Nov

Clinical fMRI applications (guest lecture: Jim Voyvodic)

Lab: Project analysis

Week 14

13. TU 24 Nov

Lab: Project analysis

Week 15

14. TH 4 Dec

Midterm II

Student group results presentations

Write-up of project due by 11:59 pm on Tuesday, December 8, 2026.